

ating a large number of windows, grids and mathematical functions on the screen of the oscilloscope." These windows can be resized as per requirements. He adds, "For instance, some people prefer having very big measurement windows and a small waveform window, and vice-versa."

Also, for ease of use, the actual vertical scale is printed on the oscilloscope's screen. Normally, if the user has switched the scale to one volt/division and the waveform is seen across three divisions each from the central axis, it is interpreted as six volts by looking at the divisions and scaling, informs Bhatia. He says, "With scaling numbers printed on the screen in time and voltage, it makes it quick and easy for the user to interpret data." Such advanced and powerful OSes are slowly trickling down to low-end oscilloscopes as well.

### Use digital trigger for precise measurements

The trigger is a key element in the scope. It allows capturing events of interest to provide a detailed analysis and ensuring a stable display for repeating signals. Traditional digital oscilloscopes implement analogue triggers, which means acquisition and trigger paths are different.

"The industry's first digital trigger is implemented by Rohde & Schwarz in their RTO and RTE series of scopes. In a digital trigger, the trigger signal is derived from samples of analogue-to-digital (A/D) converters. Therefore the digital trigger processes an identical signal that is acquired and displayed," shares Srinivasa Rao Appalla, manager, product support and applications, Rohde & Schwarz. He adds, "The digital trigger offers very low trigger jitter in real-time, combined with high-acquisition and analysis rates in RTO and RTE scopes. It enables precise measurements due to high trigger sensitivity at full bandwidth and adjustable digital filter for the signals."

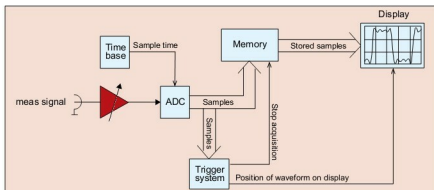


Fig. 1: Block diagram of digital trigger implementation

Fig. 1 shows a block diagram of digital trigger implementation.

### Use advanced maths operators for better power analysis

Modern oscilloscopes have powerful analysis and software capabilities; in particular, they have a lot of maths functions. Add, subtract, multiply, divide, differentiate, integrate, fast Fourier transform (FFT), square-root, absolute value, logarithms, filters and many more transforms and operators are available. You can apply these operators on captured data, and it shows you the result not only in stop mode but real-time as well. Some oscilloscopes also allow users to enter and edit formulae to actively perform advanced calculations. It is practical in various applications such as power analysis by using operators (such as integral to find out power consumption of a microcontroller).

### Some other useful features and suggestions

Here are some extra features and suggestions that you might find to be helpful.

**Making the most of triggers.** The time-out-trigger mode can help with the challenging task of capturing signals that contain different pulse widths. Also, the slope-trigger function in an oscilloscope, which triggers on the positive or negative slope of a specified time interval, is useful for capturing saw-tooth and triangular-shaped signals.

**Directly manipulate with fingers.** "In our 6000X series scope that offer bandwidth starting from 1GHz up to 6GHz, we have moved to a capacitive-enabled, multi-touch gesture-enabled display that provides similar capabilities of modern-day smartphones and tablets," informs Bhatia. For example, a user can use the multi-touch feature to effectively change the horizontal or vertical scaling by using two fingers and gesture for zooming out or in and even drag a waveform around.

**Consider five times the measured signal's bandwidth.** Correct sampling is essential for making accurate measurements and it also completely eliminates errors from your design. A general rule for analogue signals is considering five times the bandwidth of the signal you wish to measure. Coming to high-frequency signal elements, it would be advisable to set your scope to attain five to ten times over sampling.

**Make the most of those probes.** Also, appropriate probing and comprehending the correct use of ground reference and differential signals is important to go error-free. If your data lines are not grounded, make necessary adjustments, keeping in mind the effect of ground loops and noise on measurements. Make full use of probe techniques and advanced noise-cancelling features on your oscilloscope to curb noise. If required, utilise differential probes for superior measurement quality. ●